A Self-Supervised Denoising Network for Satellite-Airborne-Ground Hyperspectral Imagery

Xinyu Wang, *Member, IEEE*, Zhaozhi Luo, Wenqing Li, Xin Hu, Liangpei Zhang, *Fellow, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— Hyperspectral images (HSIs) are inevitably corrupted with various types of noise, which seriously degrades the data quality and usability. Denoising is an essential preprocessing task of HSI processing. Recently, benefiting from the great learning ability of deep learning, convolutional neural network (CNN) denoisers have obtained state-of-the-art performances for Gaussian noise removal. However, one central problem remains largely unsolved: how to deal with the complicated noise in the real-world HSIs, especially when a paired training data set is unavailable. In this article, a self-supervised hyperspectral image denoising network (SHDN) is proposed, which consists of a noise estimator and a CNN denoiser. Rather than defining a complex noise model to generate training pairs on the clean HSIs, a self-supervised training scheme is first proposed by considering the noisy HSI itself as the training data. Through the noise estimator, the realistic noise samples can be extracted and combined with the clean bands to make up the training pairs. In addition, to jointly restore the target noisy band and to maintain the spectral consistency, a flexible multi-to-single band convolutional network is designed, where the noisy band and the neighboring bands are jointly aggregated via multiscale contextualized dilated blocks and the spectral–spatial convolutional unit. Experiments on HSIs from spaceborne, airborne, unmanned aerial vehicle (UAV)-borne, and ground-based data sets demonstrate the applicability and the generalization of SHDN in the real scenarios. Additionally, the usability of the noisy bands and the suitability of the SHDN framework in the subsequent applications are verified in the land-cover mapping experiments.**

***Index Terms*— Deep convolutional neural network (CNN), hyperspectral remote sensing image, mixed noise removal, self-supervised training.**

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Xinyu Wang and Zhaozhi Luo are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: wangxinyu@whu.edu.cn; luozhaozhi2016@whu.edu.cn).

Wenqing Li is with the Cloud and Smart Industries Group, Tencent, Shenzhen 518000, China (e-mail: virginiali@tencent.com).

Xin Hu, Liangpei Zhang, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan 430072, China, and also with the Hubei Provincial Engineering Research Center of Natural Resources Remote Sensing Monitoring, Wuhan University, Wuhan 430072, China (e-mail: whu_huxin@whu.edu.cn; zlp62@whu.edu.cn; zhongyanfei@whu.edu.cn).

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I. Introduction

THE increasing availability of hyperspectral images (HSIs) obtained from diverse platforms (e.g., satellite, manned or unmanned aircraft, and ground-based platforms) provides us with the potential to observe the Earth's surface at different scales, both locally and globally [1], [2]. The rich spectral information of HSIs makes it possible to observe the Earth's surface in a wide range of remote sensing applications, such as dynamic monitoring of the environment [3], water inversion [4], anomaly detection [5], and precision agriculture [6]. Unfortunately, due to the limitations of the sensors, HSIs are inevitably contaminated by instrument noise, which is made up of thermal noise, quantization noise, and shot noise [2]. In addition, atmospheric effects may also induce noise during the imaging process, especially for spaceborne and airborne hyperspectral systems [7]. The mixed noise degradation not only severely downgrades the visual quality of HSIs, but it also limits the image interpretation in further applications [8], [9]. For example, it has been verified that each spectral band has its own unique characteristics in land-cover mapping, and the noisy bands directly impact the classification accuracy [6]. Therefore, denoising is a critical preprocessing task for HSIs, with the aim being to restore latent high-quality images from the low-quality observations [10].

For HSI denoising, the simplest approach is to directly perform the image denoising on each band individually [11]. However, using this type of approach, the additional spectral properties of HSIs are more or less ignored, which can lead to undesirable spectral or spatial distortion in the denoising results. Therefore, more effective methods have been developed by considering the unique characteristics of HSIs, i.e., the high correlations in the spectral and spatial domains. These approaches can be broadly categorized into three main groups, namely, (1) filtering-based methods, (2) optimization-based methods, and (3) deep learning-based methods.

Most of the early hyperspectral denoising algorithms fall into the filtering-based group category that separates the clean components from the noisy ones in a transformed domain, such as principal component analysis [12], [13], frequency transform [14], and wavelet transform [14], [15]. Among the different methods, the wavelet and its variations are powerful representations that have been widely used. For instance, the hybrid spatial–spectral derivative-domain wavelet shrinkage noise removal method utilizes the nonregularity of

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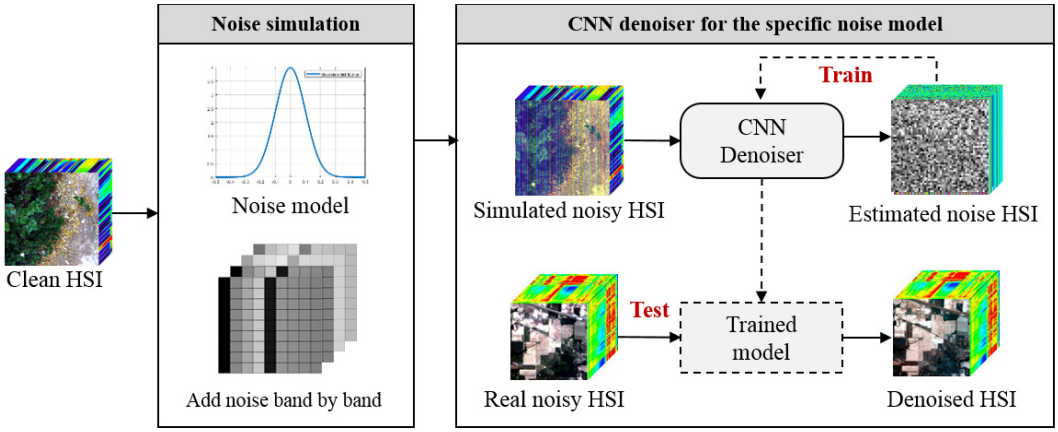


Fig. 1. General paradigm of the CNN denoiser.

HSIs in the spatial and spectral dimensions [15]. In addition, an extension of the Wiener filter, termed the multidimensional Wiener filter [14], was designed to adapt to the varying noise levels in the different spectral bands. Although these filters can effectively remove most of the noise corruption, the denoising results are easily affected by the oversmoothing effect, and the filters are inherently limited by the preselected transform basis.

Over the past decade, optimization-based HSI denoising models have become a major trend, due to their effectiveness and flexibility. The main idea of this type of methods is to restore the latent clean HSI from the noisy one via an optimization framework [16]. Specific spectral and/or spatial priors, including total variation [17], [18], sparsity [19], [20], and low-rank priors [16], [21], [22], have been adopted to make the inverse optimization problem well posed. Total variation is one of the classic image priors for preserving sharp edges during the denoising process [23]. And it has been extended to the HSI denoising task by considering the additional spectral information, in methods such as the spectral–spatial adaptive hyperspectral total variation model [17], the spatio–spectral total variation model [18], and the spectrally adaptive multidimensional nonlocal total variation model [10]. From the sparse representation perspective, the clean HSI is assumed to be a sparse linear combination of atoms from the redundant dictionary [20]. For example, Li *et al.* [19] jointly coded the intraband structure and the interband correlation of HSIs into a single sparse representation framework, and removed the noise in each spectral group adaptively. From another perspective, the highly correlated spectral–spatial features of HSIs can be modeled as the low-rank property, with the assumption that the clean HSI lies in a low-dimensional subspace of the high-dimensional image [24]. On the basis of the low-rank assumption, 2-D low-rank matrix recovery (LRMR) models [16], [25] and 3-D low-rank tensor approximation (LRTA) models [26]–[32] have been proposed and have achieved inspiring results. Generally speaking, the use of more image priors can more accurately describe the intrinsic HSI features, and thus multiple image priors can be jointly utilized under a specific optimization framework for the mixed noise removal of HSIs [21], [22]. However, the optimization-based models are more or less restricted by the hand-crafted image priors, and they cannot easily model multitype mixed noise. In addition, a better denoising performance is usually unexpectedly accompanied by a high computational burden in the optimization process.

As an alternative approach, convolutional neural networks (CNNs) have been introduced into the hyperspectral denoising task, with the aim being to automatically learn image priors from numerous noisy-clean training pairs via the end-to-end networks [33]–[36]. The general denoising paradigm is presented in Fig. 1, where the noisy images are generated from the clean HSIs according to a predefined noise model such as a Gaussian noise distribution. For example, Xie and Li [33] first utilized a deep CNN with trainable nonlinearity functions to remove the Gaussian noise in HSIs; Chang *et al.* [34] further proposed HSI-DeNet by extending the 2-D convolution into multichannel 2-D convolution, to jointly input the adjacent spectral bands and keep the spectral consistency of the HSIs; Yuan *et al.* [35] employed a spatial–spectral deep residual CNN that integrates 3-D and 2-D convolutions for Gaussian noise removal. Maffei *et al.* [36] proposed single denoising CNN (SDeCNN) for Gaussian noise removal, which allows the network to be faster without losing denoising performance by a down-sampling layer and takes a noise-level map as input for controlling the tradeoff between denoising performance and detail preservation. Benefiting from the powerful nonlinear fitting ability of CNNs, the external information from large data sets, and the fast training speed on a modern GPU, CNN-based methods have yielded a great breakthrough in the accuracy and speed of specific HSI denoising tasks [33]. However, the success of this type of method is highly dependent on a large training data set consisting of noisy-clean HSI pairs, which are often unavailable in the real-world. On the one hand, the trained model is limited to the model-driven noise simulation process and may generalize poorly to the real HSI with more sophisticated noise distributions. On the other hand, the whole-band approach may lead to an undesirable effect on the noise-free bands when only a few bands are degraded. In general, the noise degradation only occurs in

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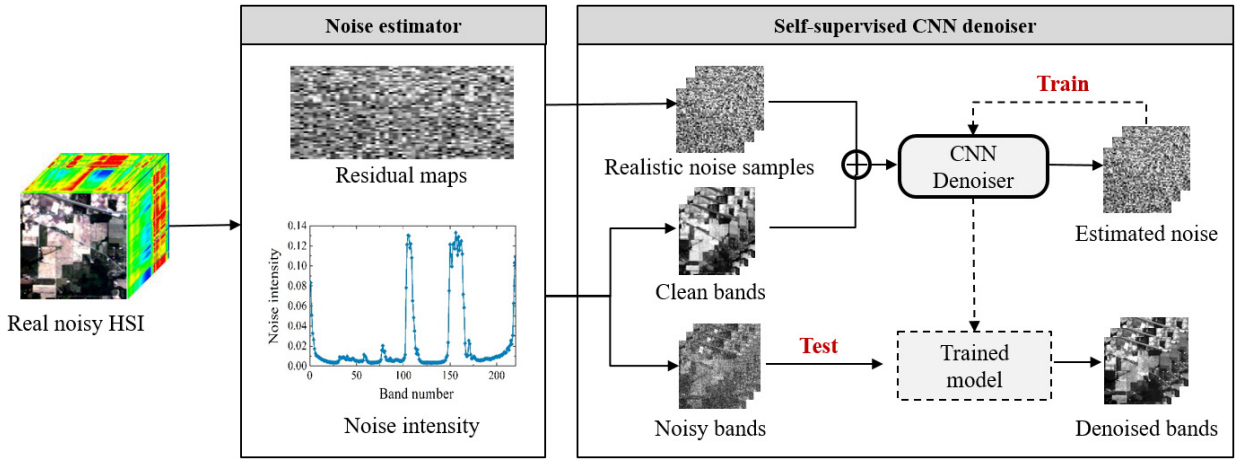


Fig. 2. Overview of the proposed SHDN framework.

a minority of the spectral bands, primarily due to the low signal-to-noise ratio caused by the optical dispersion characteristics of the spectrograph, and the noise intensity are varying among different bands, so that the whole-band assumption is not practical [7], [37].

To tackle these issues, a self-supervised hyperspectral image denoising network (SHDN) is proposed in this article, in which both the noise modeling and the training of the CNN denoiser are executed in a self-supervised manner. The SHDN framework consists of two modules, i.e., a noise estimator and a CNN denoiser. To be more specific, the noise estimator is employed to fully exploit the information of both the noise-free bands and the noisy bands, and to extract the potential realistic noise samples, including Gaussian noise, impulse noise, stripe noise, and other unknown types of additive noise. The estimated noise samples and the clean bands from a single HSI itself are then combined to build the paired training data set for the CNN denoiser. Notably, both the noise samples and the clean patches are generated from the noisy HSI itself, without external data sets, which greatly alleviates the difficulty in explicitly defining a complex noise model and building the training data set. For the architecture of the CNN denoiser, a flexible multi-to-single convolutional network is proposed on the basis of image denoising convolutional neural networks (DnCNNs), which learns to restore the target noisy band while also maintaining the spectral consistency. The noisy band (central) and the adjacent bands are jointly aggregated via the well-designed contextualized dilated spatial blocks and the spectral convolutional blocks, respectively.

The main innovations of SHDN can be summarized as follows: (1) a self-supervised denoising network is developed for the mixed-noise removal of HSIs, requiring neither a specific noise model nor hand-crafted image priors, (2) a self-supervised training scheme is first proposed by extracting the noise-clean image pairs from the noisy HSI itself via multiple regression, which overcomes the difficulty of explicitly defining a complex noise model and collecting external data sets, and (3) the intrinsic spectral–spatial feature of HSIs can be extracted via the multi-to-single CNN denoiser, which features the unique design of a multiscale contextualized dilated gradient network for the spatial denoising and multichannel spectral denoising network to maintain the spectral consistency. The extensive experiments conducted in this study showed that SHDN can obtain outstanding results, both visually and quantitatively, on HSIs obtained via multiple platforms, including Chinese Zhuhai-1 hyperspectral satellite data sets, the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Indian Pines data set, the WHU-Hi-Hanchuan unmanned aerial vehicle (UAV) data set, and the ground-based Columbia Multispectral Image Database (CAVE). Moreover, the further experiment of land-cover mapping applications on the AVIRIS Indian Pines data set verified that SHDN could effectively improve the quality and availability of HSIs. In addition, under similar conditions, a single trained model was found to generalize well on other hyperspectral scenes obtained over different geographical locations with very different land-cover types (without training) from the same sensor.

The rest of this article is organized as follows. The proposed self-supervised HSI denoising network is described in Section II. The experimental data sets from satellite-, airborne-, and ground-based platforms and the experimental results are reported in Section III, which is followed by a discussion in Section IV. And the conclusion is summarized in Section V.

II. Self-Supervised Hyperspectral Image Denoising Network

To remove the mixed noise of HSIs from multiple platforms, the SHDN framework is proposed. As illustrated in Fig. 2, the unified HSI denoising paradigm can be divided into two modules: a noise estimator and the CNN denoiser shared by both the training and the test processes. Notably, the noisy-clean training pairs can be automatically generated from the noisy HSI itself and are then used to train the CNN denoiser. Full details are provided below.

A. HSI Degradation Model

In HSIs, various types of noise commonly exist, such as Gaussian noise, impulse noise, stripe noise, and other unknown types of additive noise [16]. In this article, all the noise types are empirically considered as additive components [19], and the HSI degradation model can be formulated as follows:

$$\mathbf{y}_b = \mathbf{x}_b + \mathbf{v}_b \tag{1}$$

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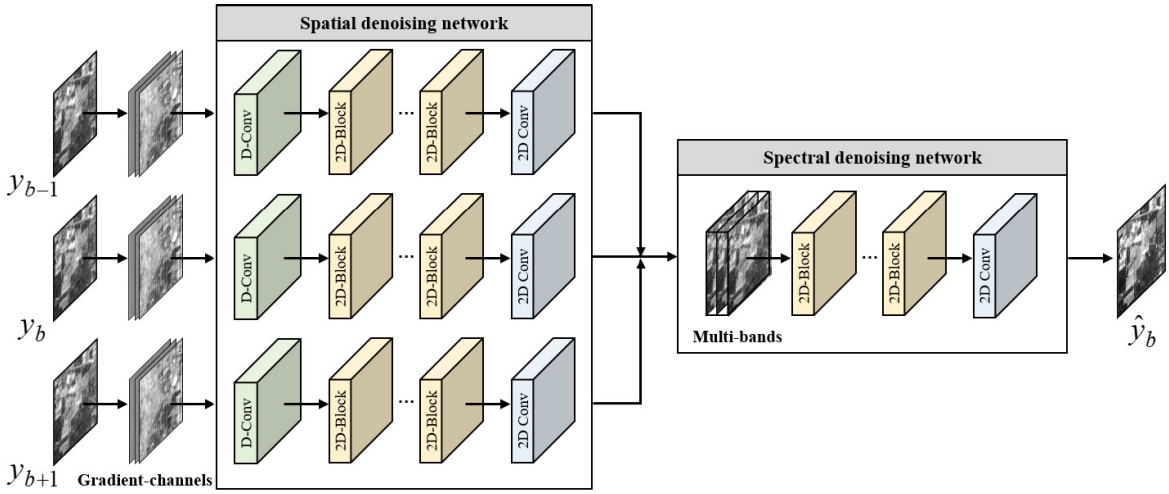


Fig. 3. Simplified architecture of the proposed multi-to-single denoising network.

where $\mathbf{y}_b \in \mathbb{R}^{M\times N}$ denotes the noisy band, $\mathbf{x}_b \in \mathbb{R}^{M\times N}$ represents the latent clean image, and $\mathbf{v}_b$ denotes all the additive noise. M and N denote the height and the width of the band, respectively. The whole HSI can be modeled as $\mathbf{Y} = \{\mathbf{y}_b\}_{b=1}^{B} \in \mathbb{R}^{M\times N\times B}$, where parameter B represents the spectral band number. Specifically, if the spectral band is noise free, the noise component $\mathbf{v}_b$ can be considered as a zero matrix. The ultimate goal of this task is to automatically distinguish and restore all the noisy bands from the noisy HSI $\mathbf{Y}$.

B. HSI Noise Estimator

To minimize the involvement of manual experts for noise modeling, the proposed SHDN framework starts with HSI noise estimation to automatically extract the potential realistic noise samples and to distinguish the high-quality clean bands. Indeed, HSI noise estimation is a fundamental processing step in many remote sensing applications, as long as they depend on a signal/noise model [38]. Over the past few decades, a series of approaches has been developed for HSI noise intensity estimation, based on the assumption that the noise is random, with a Gaussian distribution [39]–[43]. The simplest noise estimation approach is to directly use the mean of the local standard deviation (STD) values of the manually selected homogeneous blocks as the estimated noise [43]. To alleviate the need for a human operator, Gao [40] proposed the local means and local standard deviation (LMLSD) method, which divide the image into uniform blocks automatically, and employ the local statistics to represent the whole-band image. Considering the high spectral correlation in HSIs, more advanced methods have since been developed by predicting each pixel by its neighbors (spectral and/or spatial), based on multiple regression theory, and by regarding the restoration error matrix as the estimated local noise [41], [42].

In the proposed SHDN model, we combine the respective merits of multiple regression theory [38] and the statistical approach of LMLSD [40], based on a convolutional operator. The final aim is to automatically extract reliable noise samples from the noisy HSI and to distinguish the noisy bands. On the basis of the additional degradation model described in Section II-A, each pixel of the HSI can be represented by all the other spectral information via a linear function, and the prediction error is considered to be the noise. For example, if we suppose that $\boldsymbol{y}_b$ denotes the vectorized band b of size $L \times 1$, it can be expressed by the residual $B-1$ spectral bands $\boldsymbol{Y}_{\partial_b}$ of size $L\times (B-1)$ as follows:

$$\boldsymbol{y}_b = \boldsymbol{Y}_{\partial_b}\beta_b + \varepsilon_b \tag{2}$$

where β_b denotes the multiple regression coefficients of size $(B-1)\times 1$. For each band, the least square estimator of the multiple regression coefficients β_b is given by

$$\hat{\beta}_b = \left(\boldsymbol{Y}_{\partial_b}^T \boldsymbol{Y}_{\partial_b}\right)^{-1} \boldsymbol{Y}_{\partial_b}^T \boldsymbol{y}_b. \tag{3}$$

The residual error ε_b of size $L \times 1$ can be estimated as follows:

$$\hat{\varepsilon}_b = \boldsymbol{y}_b - \boldsymbol{Y}_{\partial_b}\beta_b. \tag{4}$$

Then the noise intensity can be estimated following the LMLSD statistical approach [40]. We divide all bands into the clean bands and the noisy bands according to the noise intensity threshold. For the noisy bands, the residual maps can be regarded as the mixed noise and utilized to extract the realistic noise samples. It is worth noting that the noise of specific bands after specific absorption can also be estimated by all other bands with redundant information according to multiple regression.

C. Self-Supervised CNN Denoiser

Inspired by the basic idea of DnCNNs [44], a self-supervised CNN denoiser is developed for the mixed-noise removal of HSIs. The concept of "self-supervised" refers to two aspects: the fact that the noisy-clean training pairs are generated from the noisy HSI itself via the self-supervised training scheme; and the fact that the CNN denoiser automatically learns the intrinsic image features from the training pairs, which loosens the dependence on hand-crafted image priors. The CNN denoiser contains three main components: the self-supervised training pairs, the multi-to-single band denoising network architecture design, and the spatial–spectral model training. Detailed descriptions are presented in the following.

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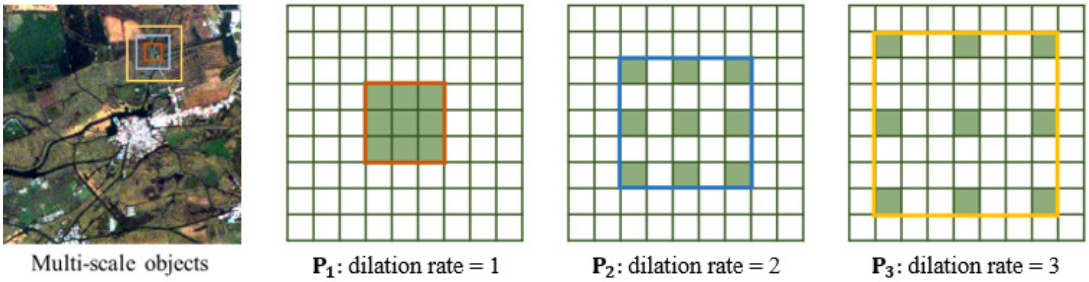


Fig. 4. Multiscale dilated convolution.

TABLE I
DETAIL ARCHITECTURE DESCRIPTION OF THE PROPOSED MULTI-TO-SINGLE BAND DENOISING NETWORK

CNN	The spatial denoising network			The spectral denoising network		
Layer	1	2-11	12	13	14-17	18
Convolutional filter	D-ConV	2D-Block	2D-ConV	2D-ConV	2D-Block	2D-ConV
Filter size	3 × 3	3 × 3	3 × 3	3 × 3	3 × 3	3 × 3
Filter number	16 × 3	64	1	64	64	1

1) Self-Supervised Training Scheme: As a data-driven method, the success of a CNN denoiser is dependent on a large training data set consisting of noisy-clean HSI pairs, which are usually unavailable in the real-world. Traditionally, most of the previous methods have used external clean hyperspectral data sets as the clean samples and have simulated the noisy HSIs from the clean ones, according to the specific noise model, such as Gaussian noise [33], [35], stripe noise [45], [46], and the combination of stripe and Gaussian noise [34]. However, such a training scheme is driven by the predefined noise model, and the trained model becomes less effective when the noise distribution does not perfectly match the noise model.

To mitigate the data dependence and to boost the model effectiveness, a self-supervised training scheme is proposed by considering the noisy HSI itself as the training data set. Based on the noise estimation results, the noisy HSI can be divided into two groups according to the noise intensity threshold: clean bands and noisy bands. Potential realistic noise samples are extracted from the estimated residual noise blocks in the noisy bands by multiple regression theory and then the paired training data set can be constructed by randomly adding the noise samples to the clean ones as follows:

$$\boldsymbol{y}_c = \mathbf{x}_c + \hat{\varepsilon}_b \tag{5}$$

where $\mathbf{x}_c$ denotes the clean bands, $\hat{\varepsilon}_b$ denotes the noise samples, and $\boldsymbol{y}_c$ denotes the corresponding noisy bands. In this way, it is very effective for the model to remove the same noise in the noise bands including specific bands. Furthermore, as imbalanced data problems may occur when the noise samples and clean ones are not equal, the smaller ones will randomly be repeated and augmented. In comparison with other CNN denoisers, the proposed self-supervised training scheme requires no external data set as auxiliary images and helps the model to approximate the real noise distribution. Such a design can not only largely alleviate the training data dependence but also provide us with a new alternative approach for related tasks where there is a lack of a paired training data set.

2) Multi-to-Single Band Denoising Network Architecture: For HSI denoising, the target noisy band is highly correlated with the neighboring spectral bands. To this end, multiple bands are usually gathered into a group and are incorporated into a multi-to-multiband network (MM) architecture [34], [45]. However, the signal and noise always dramatically vary among different spectral bands, and the noisy bands and clean bands may coexist, thus the architecture becomes less flexible.

To alleviate this issue, a multi-to-single band denoising network architecture as shown in Fig. 3 is designed on the basic architecture of DnCNN [44], to jointly learn to restore the target noisy band and maintain the spectral consistency. As shown in Fig. 4, the target noisy band (central) and the adjacent bands are gathered as a group $\{\mathbf{y}_{b-1}, \mathbf{y}_b, \mathbf{y}_{b+1}\}$, which is then input into the deep architecture, and outputs the denoising result of the target noisy band. The detailed architecture of the proposed network architecture is listed in Table I. Therein, the basic module, i.e., 2-D-block, is composed of multichannel 2-D convolution, rectified linear unit (ReLU), and batch normalization. For the spatial denoising network, the designed multiscale dilated convolutional network will be performed on each band separately. For spectral denoising, the multiple spatially denoised bands are integrated to exploit the intraband structure and the interband correlations.

For hyperspectral remote sensing imagery, the spatial sizes of the various ground objects can be quite different, so that it is difficult to capture enough contextual information from a single-scale receptive field. Therefore, the proposed multi-to-single band denoising network designed a multiscale dilated convolutional block (D-Conv) to aggregate the contextual information at different spatial scales, as illustrated in Fig. 4. The dilated convolution can exponentially enlarge the receptive field with a linearly increasing number of parameters [47], which has already been successfully used in cloud detection [48] and missing information reconstruction [49]. Based on the basic idea of dilated convolution, an improved multiscale dilated convolutional block was designed. This block

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Dataset	Platform	Sensor	Altitude (km)	Spectral range (nm)	Band number	Spectral resolution(nm)	Spatial resolution (m)
Zhuhai-1 HSI	 Zhuhai-1 **Satellite**	 OHS	500	466-940	32	2.5	10
AVIRIS Indian Pines	 ER-2 **Aircraft**	 AVIRIS	20	400-2500	220	10	20
WHU-Hi-Hanchuan	 Aibot X6 **UAV**	 Nano-hyperspec	0.25	400-1000	274	6	0.1
CAVE	**Ground** station	 Apogee Alta U260	0	400-700	31	10	9×10^{-3}

Fig. 5. Comparison of the spaceborne, airborne, UAV-borne, and ground-based HSI data sets considered in this study.

capitalizes the varying spatial context fields using dilated filters of different rates, i.e., $p = 1$, 2, and 3. With the same kernel size of 3×3, the three paths have receptive fields of 3×3, 5×5, and 7×7, respectively. For the same input, the output features concatenate the representations of the three convolution paths with different dilation factors and receptive fields to aggregate multiscale contextual information for spatially denoising. Besides, to better explore the intrinsic features, the spatial gradient along the vertical and horizontal directions will be generated as additional channels for each band image. It has been verified that the gradient channels can benefit the noise extraction, especially for directional stripe noise [45].

The spectral denoising architecture is similar with the spatial denoising module. Therein, the multiple spatially denoised bands are concatenated as input and then go through the denoising network. Specially, each spectral band will be regarded as a channel component in the 2-D convolution to exploit the intraband structure and the interband correlations. Finally, the estimated noise of the target noisy band will be the output with the residual learning block. The combination of the spatial denoising and spectral denoising can improve the spectral consistency of the denoised central band.

3) Spatial–Spectral Model Training: The average mean-square error (MSE) between the final output and the realistic noise samples extracted from the noisy HSI itself is then utilized to define the loss function

$$\text{loss}(\Theta) = \frac{1}{2n}\sum_{i=1}^{n} \| R(\mathbf{Y}_i, \Theta) - (\mathbf{y}_i - \mathbf{x}_i) \|_F^2 \tag{6}$$

where $\mathbf{Y}_i$ denotes the input of the multiple bands, comprising both the noisy band and the adjacent bands, $R(\mathbf{Y}_i, \Theta)$ denotes the output of the predicted residual noise of the central noisy band, the term $\mathbf{y}_i - \mathbf{x}_i$ is equivalent to the realistic noise in the target noisy band, n denotes the number of batches, and Θ represents the set of trainable parameters of the CNN denoiser. Then the desired clean HSI can be estimated as follows:

$$\hat{\mathbf{y}}_i = \mathbf{y}_i - R(\mathbf{Y}_i, \Theta) \tag{7}$$

where $\mathbf{y}_i$ is the target noisy band, and the term $R(\mathbf{Y}_i, \Theta)$ denotes the noise estimated by the trained model. By traversing all the noisy bands of HSI, the ideal noise-free image can be obtained. For the first band at both ends, due to the lack of the neighboring band, another highly related band will be selected as supplementary band.

III. Experiments and Analysis

In this section, the experiments are described in detail, including the multiplatform experimental data sets, the basic settings, the results obtained in both simulated noise removal and realistic noise removal, and the land-cover classification application.

A. Multiplatform Experimental Data Sets

To verify the effectiveness of the SHDN framework, four representative hyperspectral data sets from multiple platforms were employed in this study, i.e., ground-based, UAV, manned aircraft, and satellite. These data sets differed in both the spectral and spatial resolutions and were corrupted with different mixed-noise distributions. The main parameters of these hyperspectral data sets are compared in Fig. 5.

1) Chinese ZhuHai-1 Hyperspectral Satellite Data Sets: Launched in April 26, 2018, the second batch of Zhuhai-1 micro–nano satellites, including four hyperspectral satellites and a video satellite, was aimed at filling the gap of China's commercial hyperspectral satellites [50]. The project is operated by the Zhuhai Orbita Aerospace Science and Technology Corporation.[1] Operating at an altitude of 500 km, the orbita hyperspectral satellite (OHS) sensor onboard the Zhuhai-1 satellite provides 10-m spectral resolution and 2.5-nm spatial resolution HSIs, which are suitable for the accurate quantitative analysis of vegetation, water, and marine features. With a large swath width of more than 150 km, the Zhuhai-1 hyperspectral imagery is especially well suited for large-scale land-cover mapping. In this study, four Level-1B products from the Chinese Zhuhai-1 hyperspectral

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satellites were employed. The first HSI, which was acquired in Xiong'an, Hebei province of China, on October 23, 2018, was utilized to train the CNN denoiser, while the other HSIs with very different land-cover types were used for the testing. Although the radiometric calibration and atmospheric correction can improve the data quality, to some extent, there is still severe random noise at both ends of the spectral domain.

2) AVIRIS Indian Pines Airborne Data Set: The AVIRIS Indian Pines data set is a classic benchmark for denoising [17] and land-cover classification. The data were acquired on June 12, 1992, via an ER-2 aircraft. Operating at an altitude of 20 km, the acquired HSI has 220 spectral bands from 400 to 2500 nm, with a spatial resolution of 20 m [51]. After the removal of the uncalibrated bands and water absorption bands, a subimage of size $145 \times 145 \times 200$ was adopted in the experiment. The noise mainly exists at both ends of the spectrum in this image, including Gaussian noise, impulse noise, slight stripe noise, and other unknown types of additive noise.

3) WHU-Hi-Hanchuan UAV Data Set: The WHU-Hi-Hanchuan UAV data set is made up of high spectral resolution data, with the spatial resolution at the centimeter level. The data set was collected by the RSIDEA group of Wuhan University.[2] Operating at an altitude of 250 m, the Headwall Nano-Hyperspec sensor mounted on an Aibot X6 UAV obtained a flight strip with a size of 303×1217 at Hanchuan, Hubei province, China [6]. When compared with the spaceborne and airborne hyperspectral data sets, the UAV-based data set provides more detailed information about the land surface and is thus suitable for precise land-cover mapping and research on China's smallholder agriculture. Although the radiometric calibration can alleviate the inconsistency in the data, there are still minor noisy bands distributed at both ends, which are evident in the homogeneous regions.

4) Ground-Based CAVE Data Set: The ground-based CAVE [52] was acquired indoors with a cooled charge-coupled device (CCD) camera (Apogee Alta U260). The database is made up of 32 hyperspectral scenes of various real-world materials and objects, which were acquired *in situ*, ranging from 400 to 700 nm. With a fixed focus and exposure time, these *in situ* observations can be considered as noise free when compared with the spaceborne and airborne HSIs. In this study, the CAVE data set was employed for the simulated experiments.

B. Experimental Settings

For the parameters of LMLSD, the block size was set as 4×4 and the noise intensity threshold was set as 0.02. For the training data set, paired HSIs of size $40 \times 40 \times 3$ were generated from the clean bands and the estimated noise samples with a stride of 10. For the CNN denoiser, the ADAM optimizer [53] was employed with $\beta = (0.9.0.999)$ and a fixed initial learning rate of $\alpha = 0.001$ during 100 epochs. In addition, a mini-batch strategy of size 128 was utilized to boost the training procedure. The SHDN models were trained in the Python 3.6 environment with the TensorFlow package and a single NVIDIA Titan-X GPU.

To verify the effectiveness of the proposed SHDN framework, we compared it with the current state-of-the-art methods, i.e., parallel factor analysis (PARAFAC) [54], spectral domain statistics (SDS) [13], LRTA [28], LRMR [16], decomposable nonlocal tensor dictionary learning (TDL) [55], spatial–spectral total variation regularized local low-rank matrix recovery (LLRSSTV) [56], and the benchmark CNN denoiser, such as DnCNN [44] and SDeCNN [36]. Notably, the DnCNN model was trained with the original BSD 400 data set, so as to obtain the optimal results, and the SDeCNN model was trained with the Washington DC Mall data. The CNN architectures and training schemes of SHDN are described in Section V. All the algorithm code was provided by the authors, and the parameters were set by default.

C. Denoising Experiments With the Simulated Hyperspectral Data Sets

Due to the lack of atmospheric influence and other disturbances, the flower's image from the ground-based hyperspectral CAVE data set was employed as a high-quality reference for the simulated noise removal experiments. According to the real HSI degraded scenarios, three common noise types, i.e., Gaussian noise, impulse noise, and stripe noise, were selected to simulate the noise degradation. The first three bands at both ends were selected as noisy bands, and three synthetic cases were adopted as follows:

Case 1: Gaussian noise with different noise levels (the std $G \in [0, 0.1]$).

Case 2: Gaussian noise and impulse noise with different noise levels ($G \in [0, 0.1]$, $P \in [0, 0.2]$).

Case 3: Mixed noise consisting of Gaussian noise, impulse noise, and stripe noise. Based on Case 2, 50% of the noisy bands were selected to be degraded by additional stripe noise.

For a fair comparison, three quantitative indices were utilized to assess the denoising results. For the spatial dimension, the peak signal-to-noise ratio (PSNR) and the structural similarity (SSIM) indices [57] were employed to evaluate the denoising results on each noisy band, and their average values were denoted as MPSNR and MSSIM, respectively [37]. For the spectral consistency analysis, the mean spectral angle mapper (SAM) [58] was employed, which calculates the spectral fidelity between the original HSI and the denoised one. Generally, the higher SSIM and PSNR values and a lower SAM value indicate a better denoising results.

The quantitative results for all the simulated conditions, i.e., Cases 1–3, are listed in Table II. Therein, the best result is marked in bold, and the second-best performance is underlined. It can be observed that SHDN yields the best denoising performance under most conditions, especially for the mixed noise removal.

For a visual comparison, multiple representative bands in Case 3 are selected and presented in Figs. 6 and 7. Fig. 6(a) shows the noisy band 30 corrupted with a mixture of stripe noise, Gaussian noise, and impulse noise. For the denoising result of PARAFAC, it removes all the noises but at the

[2]https://www.rsidea.whu.edu.cn

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TABLE II
QUANTITATIVE EVALUATION RESULTS FOR THE SIMULATED EXPERIMENTS

	Noise level	Index	Noisy	PARAFAC	SDS	LRTA	LRMR	TDL	LLRSSTV	DnCNN	SDeCNN	SHDN
Case 1	G∈[0–0.1]	MPSNR	26.55	29.25	32.18	26.56	26.64	26.60	32.85	**37.39**	32.34	35.35
		MSSIM	0.646	0.810	0.892	0.646	0.651	0.647	0.834	**0.934**	0.881	0.928
		SAM	0.566	0.267	0.224	0.566	0.567	0.565	0.429	0.124	0.197	**0.118**
Case 2	G∈[0–0.1] P∈[0–0.2]	MPSNR	13.90	23.99	19.52	13.96	14.11	13.94	30.95	20.48	24.90	**31.68**
		MSSIM	0.232	0.614	0.434	0.232	0.244	0.234	0.828	0.399	0.693	**0.841**
		SAM	0.593	0.355	0.539	0.613	0.594	0.592	0.307	0.373	0.400	**0.179**
Case 3	*Case 2* + S∈[0–0.1]	MPSNR	13.84	23.78	19.45	13.89	14.04	13.88	27.24	20.45	24.43	**28.12**
		MSSIM	0.227	0.596	0.429	0.228	0.239	0.229	0.746	0.395	0.679	**0.753**
		SAM	0.613	0.371	0.540	0.627	0.614	0.612	0.453	0.379	0.407	**0.247**

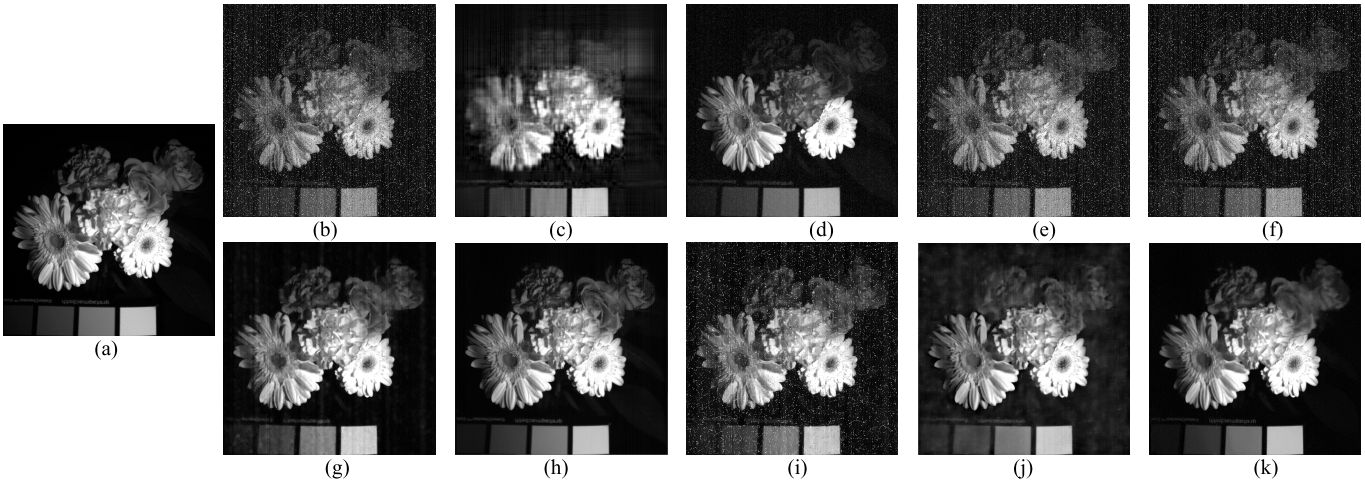


Fig. 6. Simulated random noise removal results on the flower's image from the CAVE data set (band30). (a) Original image. (b) Noisy image. (c) PARAFAC. (d) SDS. (e) LRTA. (f) LRMR. (g) TDL. (h) LLRSSTV. (i) DnCNN. (j) SDeCNN. (k) SHDN.

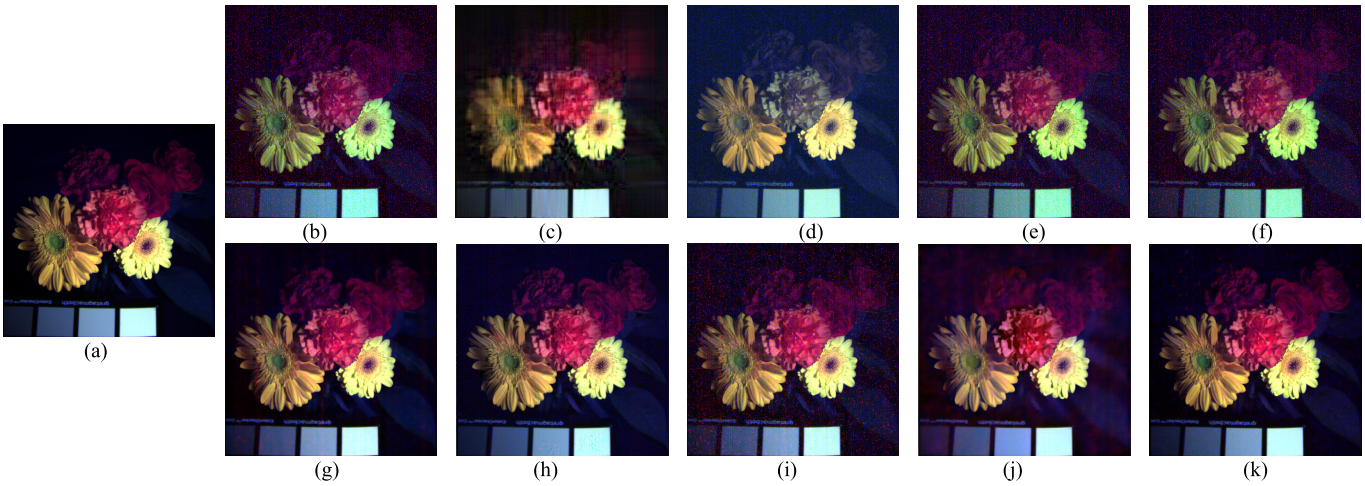


Fig. 7. Simulated random noise removal results on the flower' s image from the CAVE data set [false-color image (R: band 30, G: band 20, and B: band 1)]. (a) Original image. (b) Noisy image. (c) PARAFAC. (d) SDS. (e) LRTA. (f) LRMR. (g) TDL. (h) LLRSSTV. (i) DnCNN. (j) SDeCNN. (k) SHDN.

cost of blurring. The performances of SDS are much better in mixed noise removal for a single band. However, their performance is not stable for all the noisy bands, as presented in the false-color image. The well-known Gaussian denoiser, i.e., LRTA, TDL, DnCNN, and SDeCNN, can suppress all the Gaussian noise and alleviate the stripe noise, to some extent. However, there is still residual salt-and-pepper noise and stripes in the images. The mixed- noise model LRMR and LLRSSTV fail to remove all the noise and shows tone distortion, which can attribute to the inconsistency in the noise modeling assumption of the optimization and the realistic noise distribution in the image degradation. Comparatively, the proposed SHDN achieves a superior performance in both spatial–spectral detail preservation and mixed-noise removal.

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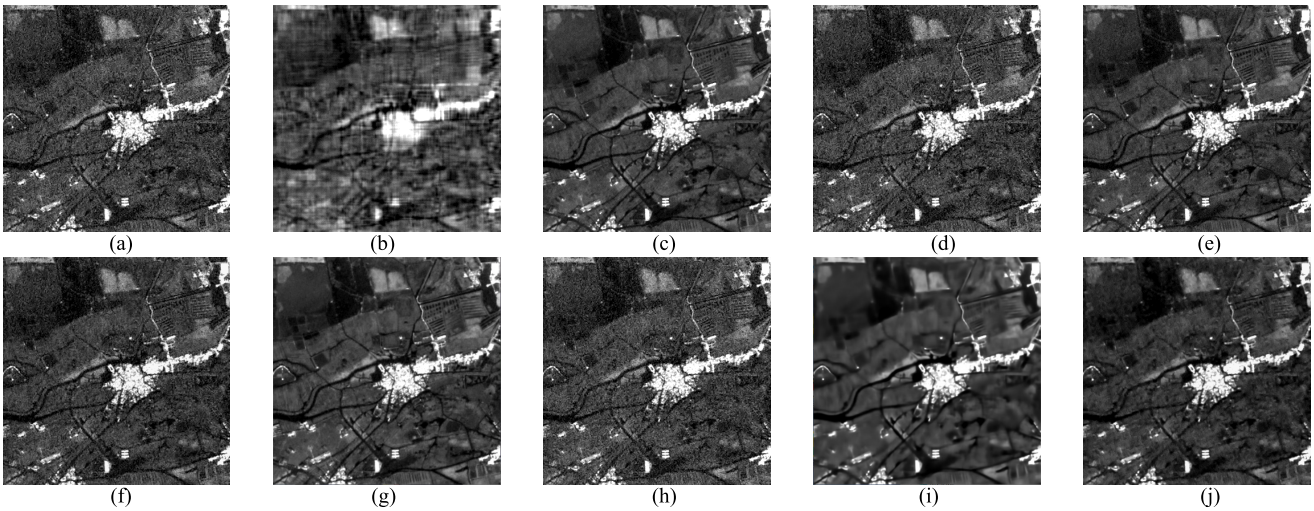


Fig. 8. Realistic noise removal results on the Zuhai-1 hyperspectral data set (band 3). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.

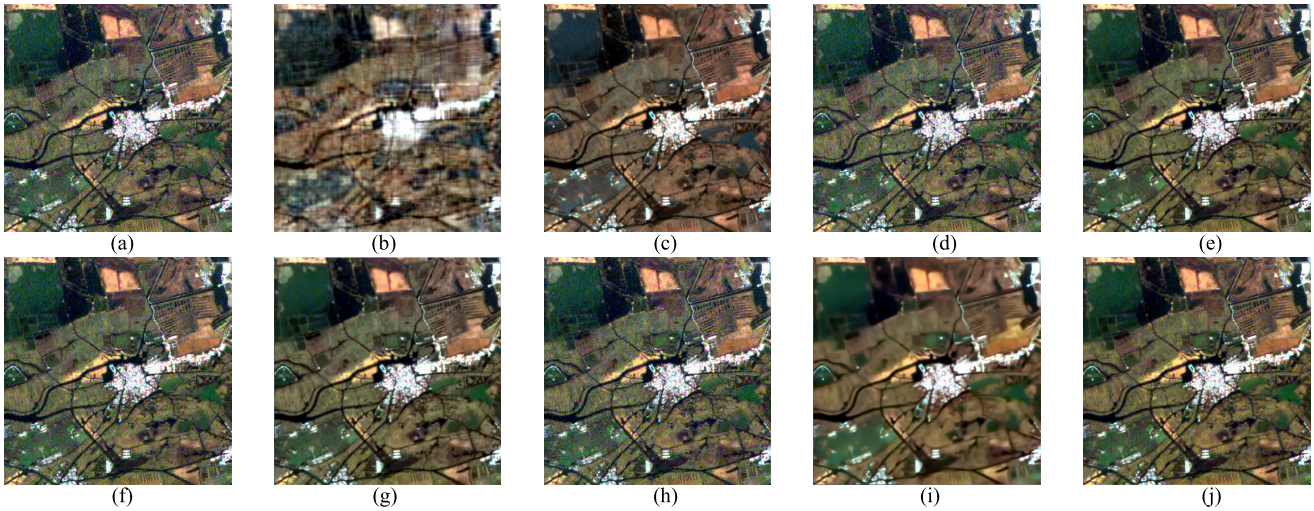


Fig. 9. Realistic noise removal results on the Zuhai-1 hyperspectral data set (R: band 16, G: band 6, and B: band 1). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.

D. Denoising Experiments With the Real Hyperspectral Data Sets

To further verify the effectiveness and practicability of the proposed SHDN framework, three hyperspectral data sets from different platforms, i.e., manned aircraft, UAV, and satellite, were adopted in the realistic mixed-noise removal experiments. Due to the absence of the ground truth (reference), the visual effects and the subsequent land-cover mapping result on the AVIRIS Indian Pines data set were adopted to compare the denoising performance. For simplicity, all the HSIs were linearly normalized to [0, 1].

1) Chinese Zhuhai-1 Hyperspectral Satellite Data Sets: The Zhuhai-1 HSI is 6282 × 6227 × 32 in size, and a subimage with a size of 400 × 400 × 32 was selected for both the training and testing. Fig. 8(a) presents the noisy band 3 as an example, which was corrupted with severe random noise. Therein, the PARAFAC method tends to introduce unexpected ringing artifacts and fails to preserve the local details. The SDS model achieved better denoising performances in the spatial domain, but some spectral distortion is unexpectedly introduced, which is evident in the false-color results shown in Fig. 9(c). The tensor-based LRTA and TDL denoising methods also failed to remove all the random noise, especially for the impulse noise. The classic CNN denoiser, i.e., DnCNN, became less effective in mixed noise removal, which can be easily explained by the fact that the realistic noise of the Zhuhai-1 hyperspectral satellite data does not totally obey the predefined Gaussian distribution assumption in the noise simulation process. The LRMR, LLRSSTV, SDeCNN, and the proposed SHDN models achieved inspiring denoising performance, both spatially and spectrally. Comparatively, the SHDN performs better in time efficiency, as listed in Table III, which indicates it is practical in real applications.

2) AVIRIS Indian Pines Data Set: The second experiment in the realistic noise removal utilized the widely used airborne AVIRIS Indian Pines data set. Figs. 10 and 11 present multiple

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TABLE III
RUNNING TIME FOR THE REAL-DATA EXPERIMENTS

Dataset	PARAFAC	SDS	LRMR	LRTA	TDL	LLRSSTV	DnCNN	SDeCNN	SHDN
Zhuhai-1 HSI	161.93	1.68	28.78	2.83	38.35	104.63	**1.56**	3.29	2.32
AVIRIS Indian Pines	1007.20	**0.33**	17.39	1.99	20.33	75.74	1.70	3.47	1.68
WHU-Hi-Hanchuan	3347.62	**5.31**	318.04	62.70	721.86	14469.43	10.08	73.36	9.97

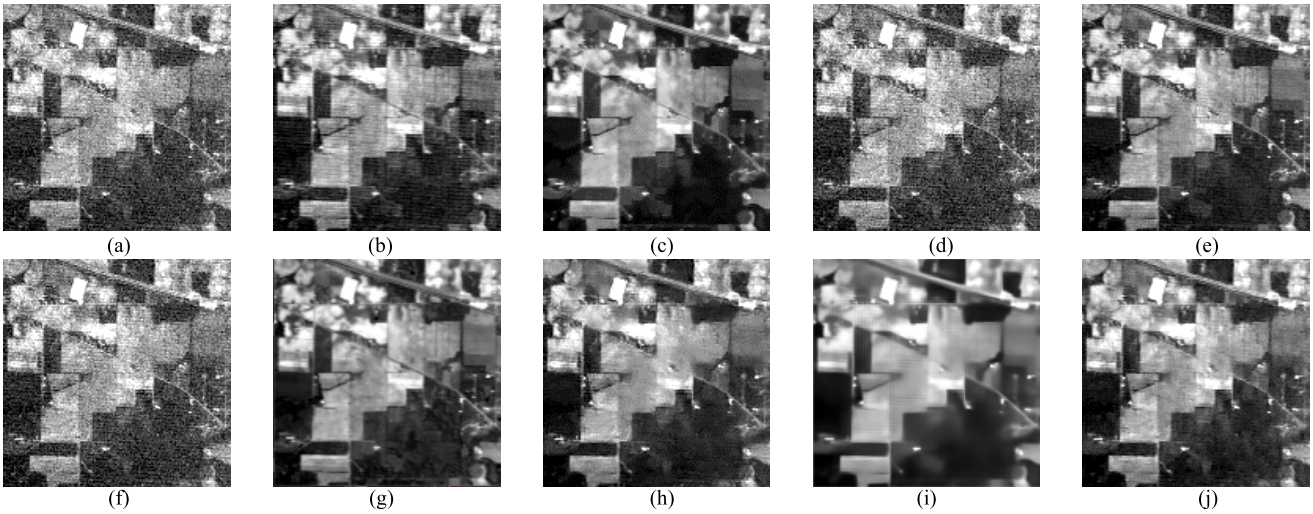


Fig. 10. Realistic noise removal results on the AVIRIS Indian Pines data set (band 3). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.

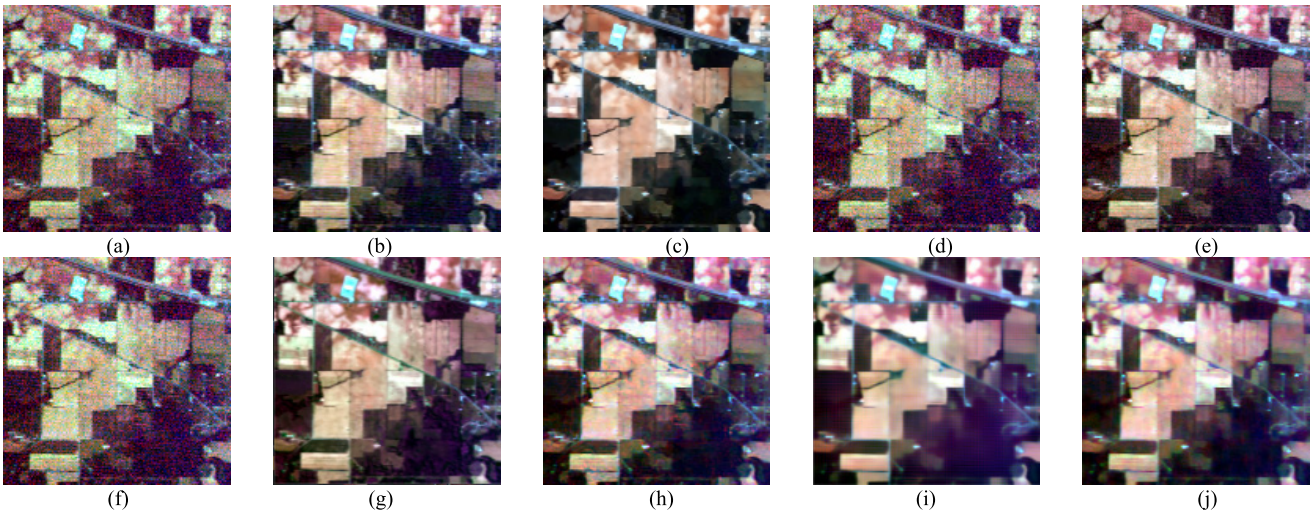


Fig. 11. Realistic noise removal results on the AVIRIS Indian Pines data set (R: band 145, G: band 24, and B: band 2). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.

noisy bands for visualization. As for the result of PARAFAC, the stripe noise still exists in the denoising result. The SDS model and SDeCNN yield a relatively better denoising performance, but the undesired banding effect is apparent. Although LRTA, TDL, and DnCNN are adept at Gaussian noise removal, all of them become less effective in the realistic mixed-noise removal of HSIs. The low-rank prior-based method LRMR struggles for a balance between noise removal and spectral–spatial information preservation, since they make a strong assumption of the low-rank constraint for the signal. However, the structural stripe noise cannot be effectively removed, as shown in the lower-right corner of Fig. 10(e). LLRSSTV and SHDN outperform all the other methods, as it removes nearly all the noise corruption and preserves most of the local details. Comparatively, LLRSSTV is time-consuming, while the SHDN shows high time efficiency as listed in Table III, which indicates the SHDN is practical in real applications.

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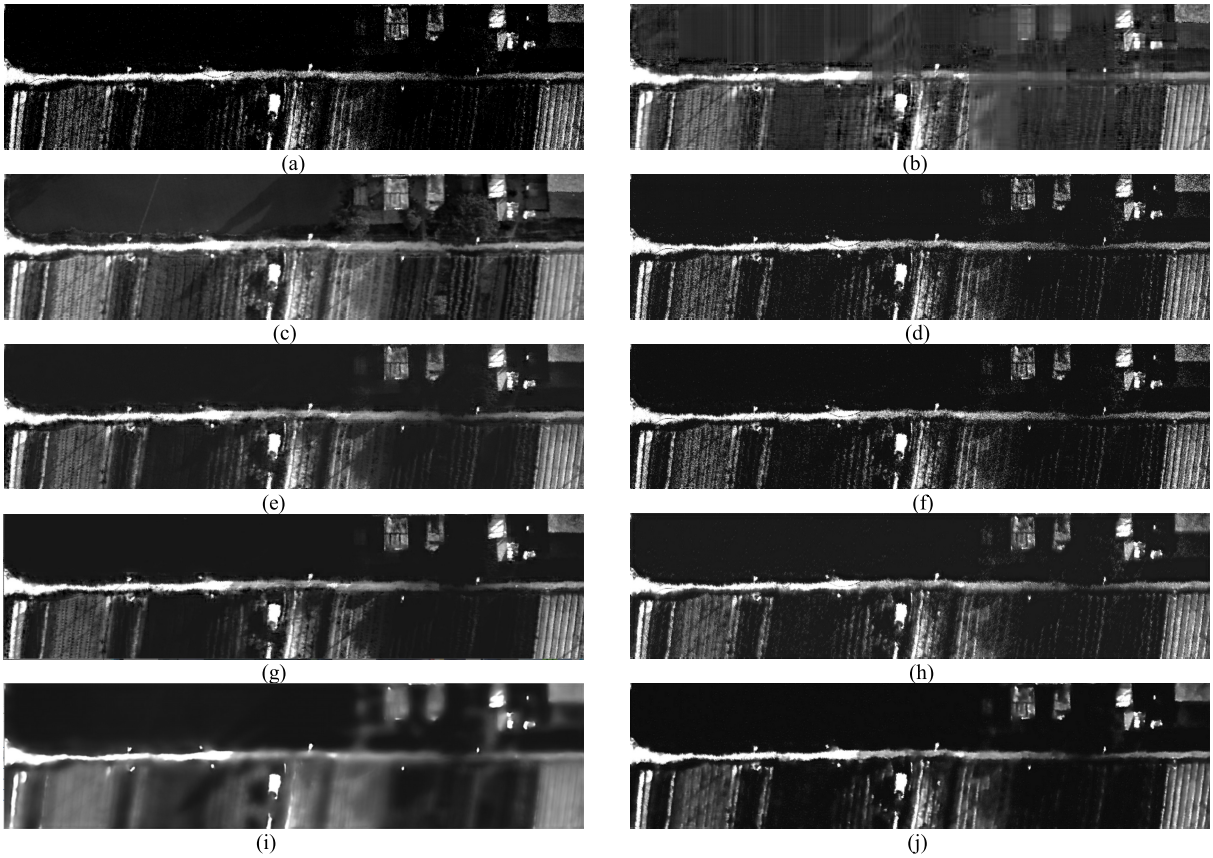


Fig. 12. Realistic noise removal results on the WHU-Hi-Hanchuan UAV data set (band 3). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.

3) WHU-Hi-Hanchuan UAV Data Set: The third experiment adopted a subimage of size $1200 \times 300 \times 270$ from the WHU-Hi-Hanchuan UAV data set [59], [60]. This data set was first used as a denoising indicator, because there were only a few bands corrupted with noise, and the noise was relatively slight, as shown in Figs. 12(a) and 13(a). Most of the denoising results are consistent with the results of the above experiments. PARAFAC fails to remove all the noise distributed in the image and introduces unexpected blurring during the denoising process, which is evident in the construction zone. The SDS method and SDeCNN remove nearly all the noise, at the cost of severe spatial distortion. For the denoising results obtained under the low-rank assumption, the results of LRMR and LRTA appear to be effective in noise removal from a visual perspective. However, some stripe-like artifacts are introduced at the boundary between water and vegetation regions, as presented in Fig. 12. LLRSSTV also performs effectively in noise removal from a visual perspective. Nevertheless, the time consumed for noise removal is unbearable, as listed in Table III. Under the Gaussian noise assumption, the TDL model and DnCNN model also became less effective for this real-world UAV data set, which demonstrates our point that the realistic noise in HSIs is more complicated than a Gaussian distribution. Comparatively, SHDN performs best in mixed-noise removal of the noisy bands, which further demonstrates the robustness and the efficiency of the SHDN framework for UAV data sets of high spatial and spectral resolutions.

E. Application to Land-Cover Mapping

Land-cover mapping at multiple scales is important for remote sensing applications, such as ecosystem health monitoring, water quality studies, and sustainable land management. In order to quantitatively assess the denoising results and analyze the usability and practicality of SHDN, additional land-cover mapping experiments were conducted on the AVIRIS Indian Pines data set. Specifically, to further assess the usability of the noisy bands, the classification results for the selected clean bands were also compared.

The 30 samples were randomly selected for training from the AVIRIS Indian Pines data set. The remaining samples of the data set were then used for testing. The land-cover results obtained by the support vector machine (SVM) classifier of the data set are presented in Fig. 14. And the land-cover classification accuracies are listed in Table IV, including the overall accuracy (OA) and the Kappa coefficient (Kappa). Benefiting from the previous denoising process, most denoising methods help to improve the classification results, which demonstrate the necessity of HSI denoising in real applications. SHDN has superior results to state-of-the-art techniques on the AVIRIS Indian Pines data set. The classification experiment indicates that the proposed SHDN

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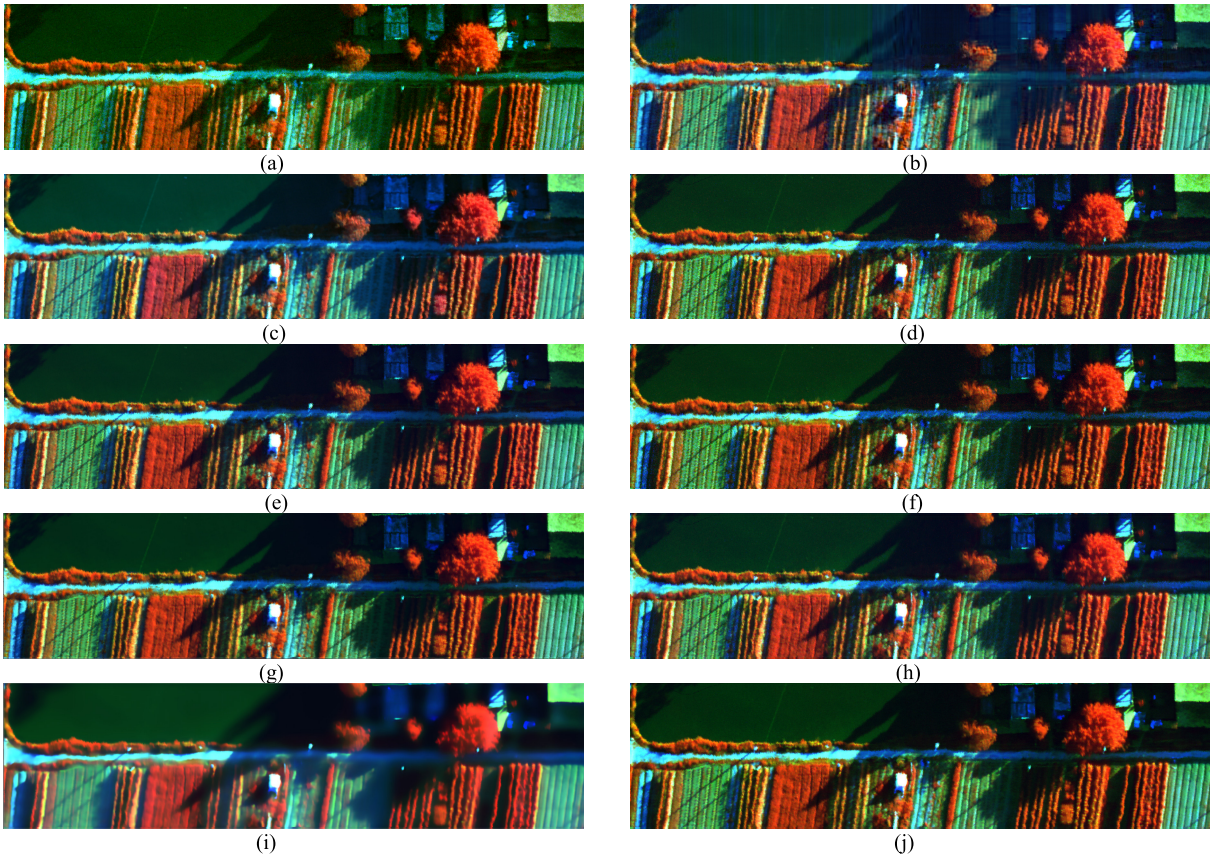


Fig. 13. Realistic noise removal results on the WHU-Hi-Hanchuan UAV data set (R: band 260, G: band 110, and B: band 3). (a) Original image. (b) PARAFAC. (c) SDS. (d) LRTA. (e) LRMR. (f) TDL. (g) LLRSSTV. (h) DnCNN. (i) SDeCNN. (j) SHDN.



Fig. 14. Land-cover mapping on the denoising results of the AVIRIS Indian Pines data set via SVM. (a) Ground truth. (b) Original image. (c) Clean bands. (d) PARAFAC. (e) SDS. (f) LRTA. (g) LRMR. (h) TDL. (i) LLRSSTV. (j) DnCNN. (k) SDeCNN. (l) SHDN.

not only greatly improves the data quality but also contributes to the success of the further application.

IV. Discussion

A. Analysis of the HSI Noise Estimation

HSI noise estimation is one of the most critical steps in denoising, which can also have a subsequent positive effect on some traditional remote sensing tasks. To analyze the accuracy of the noise estimation, the relative band importance and estimated noise level of each spectral band for the AVIRIS Indian Pines data set are shown in Fig. 15. Therein, a higher relative band importance indicates that the band plays a more significant role in the classification task, while a higher noise level denotes a more severely noise contaminated condition.

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TABLE IV
CLASSIFICATION ACCURACY RESULTS FOR THE REAL HYPERSPECTRAL DATA SETS

Dataset	Index	Original	Clean	PARAFAC	SDS	LRTA	LRMR	TDL	LLRSSTV	DnCNN	SDeCNN	SHDN
AVIRIS Indian pines	OA	66.53	73.40	66.45	69.89	66.49	72.53	67.27	77.12	75.19	81.77	79.84
	Kappa	0.623	0.699	0.622	0.662	0.623	0.690	0.631	0.740	0.720	0.794	0.771

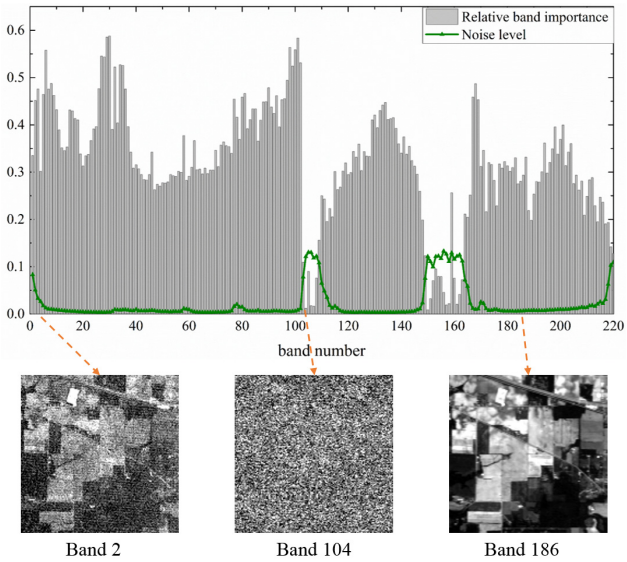


Fig. 15. Relative band importance and estimated noise level of each spectral band and some corresponding spectral bands for the AVIRIS Indian Pines data set.

It can be seen that the high-quality bands play a relatively important role in the land-cover mapping while the water absorption bands are degraded with serious noise and have little importance in the further application. As depicted in Fig. 15, the noise intensity estimated by the SHDN framework is consistent with both the visual effect and the relative band importance in classification, which verifies the accuracy of the noise estimation process used in SHDN.

B. Effectiveness of the Self-Learning Scheme

Although a larger training data set will have a positive effect on the training of the data-driven CNN denoiser, the external data set and specific noise model may greatly decrease the capacity of the trained model in practice. To address the data dependence, the proposed SHDN framework first introduces a self-learning scheme, which learns the denoiser from the noisy HSI itself by separating the spectral bands into noisy and clean ones. To verify the effectiveness of the self-learning scheme, we trained two CNN denoisers with an external data training scheme and the self-supervised training scheme, respectively. Under the same network architecture, the external learning model was trained on the ground-based CAVE data set with simulated mixed noise, including Gaussian noise, impulse noise, and stripes while the self-learning model was trained on the clean bands of the Zhuhai-1 HSIs, with the realistic noise extracted from the noisy bands. Notably, the number of training samples was set to be the same, for a fair comparison. Despite the hybrid noise assumption of the external learning model, some random noise is still apparent in the denoising result, as shown in Fig. 16(b). From the result shown

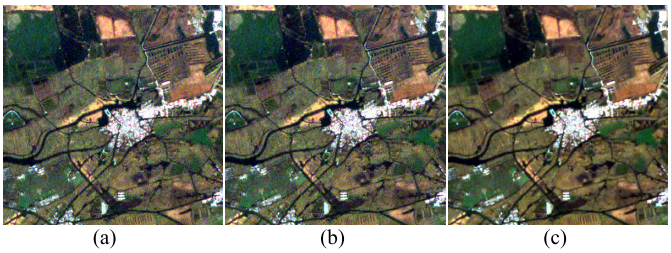


Fig. 16. Realistic noise removal results on the Zuhai-1 hyperspectral data set. (a) Original image. (b) SHDN (external data). (c) SHDN (self-supervised).

TABLE V
QUANTITATIVE EVALUATION RESULTS FOR THE CAVE HYPERSPECTRAL DATA SET WITH SIMULATED NOISE UNDER CASE 3

Index	SS	MM	MS	SHDN
MPSNR	24.79	20.98	25.54	**28.12**
MSSIM	0.638	**0.809**	0.689	0.753
SAM	0.448	0.278	0.288	**0.247**

in Fig. 16(c), it can be demonstrated that the self-learning scheme provides us with the potential for the deep CNN to learn a denoiser from the internal information of the noisy HSI itself.

C. Effectiveness of the CNN Architecture

Due to the highly correlated spectral–spatial relationship and the multiscale contextual information of HSIs, a dilated convolutional block and gradient channels are encoded into the multi-to-single band network architecture. To verify the unique design of the proposed CNN architecture, it was compared with several classic CNN architectures, including a single-to-single band network (SS), an MM, and two multi-to-single band networks with/without the contextualized dilation and gradient channels (SHDN/MS). For a fair comparison, all the CNN models were trained with the same number of training samples from the flower image of the CAVE data set. A simulated experiment under Case 3 with mixed noise degradation was conducted for the comparison. As listed in Table V, compared with the SS and MM architectures, the original MM achieves better signal preservation but fails to restore the local details, which is reflected in the higher SAM value. With the incorporation of the contextualized dilated convolution modules and the gradient channels, the proposed SHDN framework performs much better than the other MS models. Taking both the denoising effectiveness and the network flexibility into consideration, it can be seen that SHDN is a practical and competitive approach for mixed-noise removal of HSIs.

D. Generalization Analysis

Based on the self-learning scheme of SHDN, the model is trained on a single noisy HSI. To verify the generalization ability of the trained model, we used the model trained on

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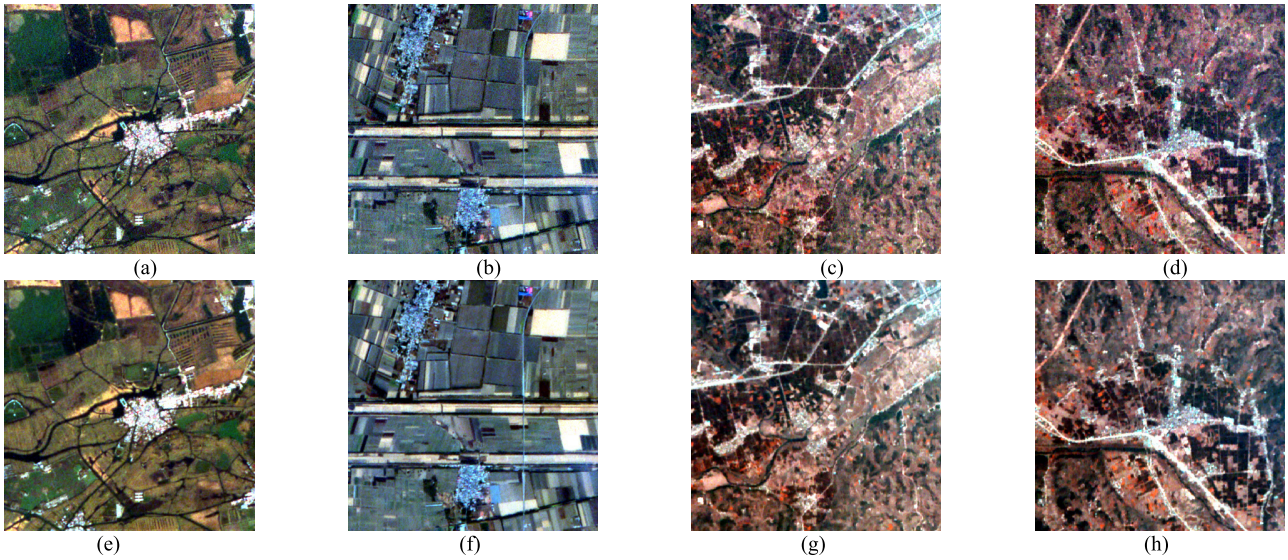


Fig. 17. Realistic noise removal results with a single trained model via SHDN on different subscenes from Zhuhai-1 hyperspectral satellite. (a) Image-1. (b) Image-2. (c) Image-3. (d) Image-4. (e) SHDN (training). (f) SHDN (test). (g) SHDN (test). (h) SHDN (test).

Image 1 from the Zhuhai-1 hyperspectral satellites and tested it on three subscenes obtained over different geographical locations, i.e., images 2–4. Clearly, most of the random noise can be effectively removed, without any textural distortion, as shown in Fig. 17. It can be demonstrated that SHDN possesses a powerful generalization ability, in that the trained model can reduce the noise in a given noisy HSI (with the training process) and can further reduce the noise in other HSIs (without the training process) obtained under similar conditions from the same sensor.

V. Conclusion

In this article, we have described how we attempted to remove mixed noise by exploring the intrinsic features of the noisy HSI itself via a deep CNN. The proposed SHDN framework consists of a noise estimator and a CNN denoiser, both of which work in a self-supervised manner. The noise estimator is employed to fully exploit the information of both the noise-free bands and the noisy bands and to extract the potential realistic noise samples, including Gaussian noise, impulse noise, stripe noise, and other unknown types of additive noise. The estimated noise samples and the clean bands can then be combined to solve the issue of building a paired training data set for the CNN denoiser. To our knowledge, this is the first attempt to explore the potential of a self-learning scheme for HSI denoising by considering the realistic noise distributions. Extensive experiments were conducted on multiplatform data sets, including Chinese Zhuhai-1 hyperspectral satellite data sets, the airborne AVIRIS Indian Pines data set, the WHU-Hi-Hanchuan UAV data set, and the ground-based CAVE data set, and showed competitive denoising performance. Despite the encouraging performance of SHDN, still some limitations are to be further investigated. One limitation of the proposed model is that fixed-pattern structural noise, such as the complex types of densely distributed stripe noise in spaceborne HSIs, is difficult to extract and estimate. Furthermore, to further improve the generality of the framework, the usability of the noisy bands and the denoised bands should be systematically investigated, in classical HSI applications such as change and anomaly detection, unmixing, and quantitative retrieval.

Acknowledgment

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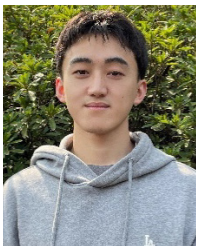
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Xinyu Wang (Member, IEEE) received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University in 2019.

Since 2019, he has been an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His major research interests range over hyperspectral unmixing, anomaly detection, and restoration.



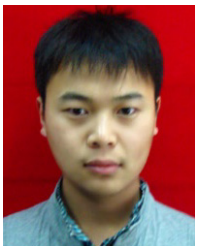
Zhaozhi Luo received the B.S. degree in remote sensing science and technology from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2020, where he is pursuing the M.S. degree in resources and environmental engineering.

His major research interests include hyperspectral image restoration and unsupervised learning.



Wenqing Li received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2017, and the M.S. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, in 2020.

She is working with the Cloud and Smart Industries Group, Tencent, Shenzhen, China. Her major research interests include hyperspectral image restoration, machine/deep learning and big data analysis.



Xin Hu received the B.S. degree in surveying and mapping engineering from Northeastern University, Shenyang, China, in 2017. He is pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China.

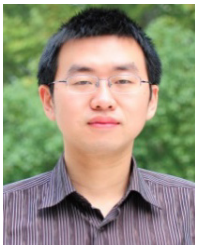
His major research interests include agricultural remote sensing and hyperspectral image classification.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He was a Principal Scientist for the China State Key Basic Research Project from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He is a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He has authored more than 700 research articles and five books. He is the Institute for Scientific Information (ISI) highly cited author. He holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a Fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He also serves as an Associate Editor or an Editor for more than ten international journals. He is serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is the Founding Chair of IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis, and Applications of Remote Sensing (RSIDEA) Research Group. He has authored more than 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a Fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE GRSS Data Fusion Contest and the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.

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